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Al-Balqa' Applied University / Faculty of Engineering Technology
General Chemistry / Mid-Term Exam

Time: 90 minutes **First Semester (2018/2019)** **Date: 15/ 11/2018**

Physics and Basic Sciences Dept.

Section: (١ - ١٢) **Instructor:** د. ريم ناصر

Student Name: [redacted]

Question No.	1	2	3	4	5	6	7	8	9	10
Answer	a	d	b	c	b E	a D	c	c A	e	d
Question No.	**11**	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
Answer	b	e	a C	e	e	b	a	B d	a E	e

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Useful Data: Avogadro's No. = 6.022x 10^{23}

Q1: Circle the most correct answer in the following questions:

1. The set of quantum numbers, $n = 4$, $l = 3$, $m_l = 2$ is describing an electron in a…

(a.) 4f orbital b. 3p orbital c.it is not allowed d. 4d orbital e. 4s orbital

2. Which of the following sets of quantum numbers is **not** allowed?

a. $n = 3$, $l = 1$, $m_l = -1$, $m_s = +½$ b. $n = 2$, $l = 1$, $m_l = 0$, $m_s = +½$

c. $n = 3$, $l = 0$, $m_l = 0$, $m_s = +½$ d. $n = 2$, $l = 2$, $m_l = -1$, $m_s = -½$ e. all are allowed

3. How many moles of hydrogen atoms exists in 0.01 mole of $(NH_4)_2HPO_4$?.

a. 0.01 b. 0.09 c. 0.5 d. 0.08 e. 0.02

4. How many grams of N_2O are formed, if 0.46 mole of NH_4NO_3 is used in the reaction? Given the **unbalanced chemical equation.** $NH_4NO_{3(s)} \rightarrow N_2O_{(g)} + H_2O_{(l)}$

a. 40 g b. 80 g c. 20 g d. 10 g e. 16g

5. Consider the reaction: $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$

If 0.86 mole of MnO_2 and 48.2 g of HCl react, how many grams of Cl_2 will be produced?

a. 46.8 b. 11.7 c. 64.2 d. 18.5 e. 23.4

6. Lithium reacts with nitrogen when heated to form lithium nitride according to the following balanced equation: $6Li(s) + N_2(g) \rightarrow 2Li_3N(s)$

If 5.89 g of Li_3N is formed when 12.3 g of Li are heated with 33.6 g of N_2, what is the percent yield of the reaction? a. 14.3 % b. 48.2% c. 34.2% d. 28.6% e. 57.2%

7. What is the percent water in hydrated calcium chloride $CaCl_2.2H_2O$?

a. 66.67% b. 32.47% c. 24.51% d. 12.26% e. 16.23%

8. Calculate the minimum sum of all coefficients in the **balanced** chemical reaction:

$Fe_2O_3 + HNO_3 \rightarrow Fe(NO_3)_3 + H_2O$

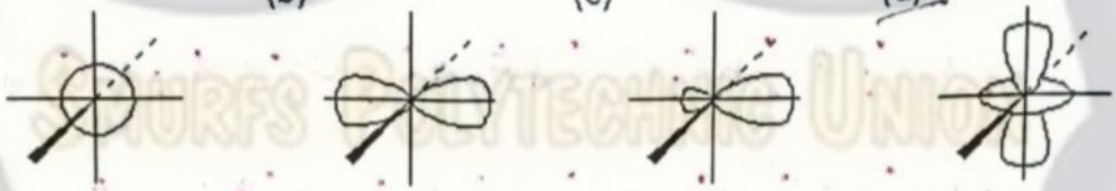
a. 12 b. 8 c. 6 d. 15 e. 18

9. Which of the following ions has the smallest diameter?

a. O^{2-} b. Na^+ c. F^- d. Ne e. Al^{3+}

10. The shape of one of the atomic 3d orbitals is...

(a) (b) (c) (d)



11. Choose the electron configuration of the zinc ($_{30}Zn$) atom (in its normal state):

a. $[Ar]4s^2\ 4p^6$ b. $[Ar]4s^2\ 3d^{10}$ c. $[Ar]4s^1\ 4d^{10}$ d. $[Kr]5s^2\ 5d^{10}$ e. $[Kr]5s^1\ 5d^{10}$

12. How many orbitals are there on the 3rd energetic level?

a. 16 b. 2 c. 3 d. 4 e. 9

13. Atomic radii of chemical elements in a **group increase** from:

a. left to right b. right to left c. top to bottom d. bottom to top. e. stays constant

14. Which of the following electron configurations represents the electron configuration for a magnesium cation $_{12}Mg^{2+}$?

a. $1s^2 2s^2 2p^6 3s^2$ b. $1s^2 2s^2 2p^6 3s^2 3p^2$ c. $1s^2 2s^2 2p^4$ d. $1s^2 2s^2 2p^5$ e. $1s^2 2s^2 2p^6$

15. Which of the following species has the *most* unpaired electrons S^+, S^{2+}, S, or S^- ?

a. S b. S^- c. S^+ d. S^{2+} e. all have the same number of unpaired electrones.

16. A substance (X) containing C, H, N, and O. Given the following percent composition by mass: 19.8% C, 2.5% H, 11.6% N. What is the molecular formula given that the mass of 0.01 mole of X is weighing 1.21 g?

a. $C_4H_6N_2O_{10}$ b. $C_2H_3NO_5$ c. $C_3H_3NO_4$ d. $C_4H_3N_2O_5$ e. $C_5H_5NO_3$

17. Calculate the number of N, in 1.68×10^4 g of urea $[(NH_2)_2CO]$.

a. 3.37×10^{26} b. 3.37×10^{23} c. 6.74×10^{26} d. 6.7×10^{25} e. 1.19×10^{26}

18. Which element of the following has the highest electron affinity?

a. $_{20}Ca$ b. $_{35}Br$ c. $_{33}As$ d. $_{34}Se$ e. $_{36}Kr$

19. Which property is identical for elements from the same period?

a. atomic radius b. ionization energy c. the number of valence electron
d. chemical properties e. None is correct

20. Which element would have the **greatest difference** between the first ionization energy (I_1) and the second ionization energy (I_2)? a. $_{19}K$ b. $_{20}Ca$ c. $_{13}Al$ d. $_{14}Si$ e. $_3Li$